Test Report I

LB.030.28/14-177-VT01/I

DGUV Test
Prüf- und Zertifizierungsstelle
Elektrotechnik
Fachbereich Energie Textil
Elektro Medienerzeugnisse

Client and Manufacturer:	ASSA ABLOY Entrance Systems Albany Door Systems GmbH Am Mondschein 25 59557 Lippstadt, Germany
Test object:	Passive laser protection walls
Type designation:	See Attachment 1
Date of manufacture:	2014
Test specifications:	Test specification PL-Info 007 from 05 Dec. 2013[1)] — Passive laser protection walls – Determination of protection duration for application-specific uses DIN EN 60825-4:2011-12 Annex D — Safety of laser products – Part 4: Laser guards
Test location:	TRUMPF Laser- und Systemtechnik GmbH, 71254 Ditzingen, Germany
Testing dates:	06 Nov. 2014
Tests performed and test results:	See Attachment (Test Protocol from 12 May 2015)
Note:	The EC Machinery Directive as well as DIN EN ISO 11553-1 *Laser processing machines* obligate the manufacturer of a laser processing machine to carry out a risk assessment. It must be taken into consideration that this Test Report is limited merely to an evaluation of the protective action of passive laser protection walls while the manufacturer must undertake a comprehensive assessment of the laser safety concept.
Remarks:	[1)] This Test specification does not fall within the scope of accreditation.
Total page count:	2 + 1 attachment The Test Report may be rendered only in its unabridged form.
Date of issue:	12 May 2015
Signatures:	Malte Gomolka Test engineer Martin Mehlem Laboratory supervisor

General information and explanations

1. Test specification PL-Info 007 from 05 Dec. 2013 substantiates and specifies the normative Test methods according to DIN EN 60825-4:2011-12 Annex D, as neither reproducible nor explicit conditions for testing are prescribed herein. The Test specification does not fall within the scope of accreditation.

2. An individual examination of measurement uncertainty while determining the duration of protection was not performed. Measurement uncertainties (e.g. time measurement, test setup, sample pre-treatment, laser conditions, etc.) are, in fact, taken into account in the 0.7 safety factor.

3. The test results are identical to the test results from 15 Oct. 2013 (File reference 13-166, Attachment 6, Trial A).

Please feel free to contact us if any further information is required (Tel. +49 (0)221 3778-6310 or gomolka.malte@bgetem.de).

Attachment
Test Protocol from 12 May 2015

DGUV Test Prüf- und Zertifizierungsstelle Elektrotechnik Fachbereich Energie Textil Elektro Medienerzeugnisse	Test protocol No. 1 Laser protection wall	Az.: LB.030.28/14-177-VT01/I Page: 1 of 2 Name: Gomolka Date: 12 May 2015

Client: ASSA ABLOY Entrance Systems
Albany Door Systems GmbH
Am Mondschein 25
59557 Lippstadt, Germany

Participants: Mr. Frede (ASSA), Mr. Jegel (Trumpf) and Mr. Gomolka (Electrical engineering testing and certification facility in DGUV Test)

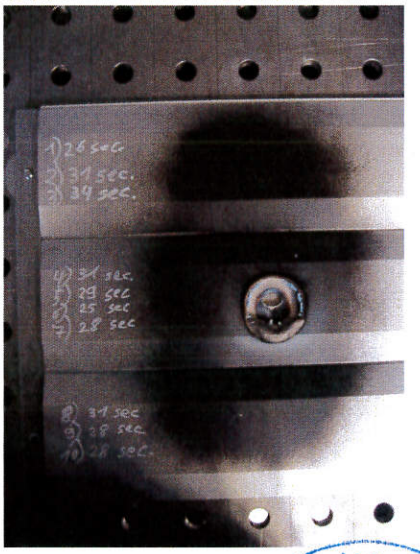
Test location: TRUMPF Laser- und Systemtechnik GmbH, 71254 Ditzingen, Germany

Test specification(s): Test specification PL-Info 007 from 05 Dec. 2013 (does not fall within the scope of accreditation) and DIN EN 60825-4:2011-12

Laser specifications:

- Manufacturer: TRUMPF TruDisk 8002
- Wavelength: 1030 nm
- Optics: TRUMPF PFO 3D
- Focal length: 450 mm
- Fibre core diameter: 0.2 mm
- Focus diameter: 0.65 mm
- Raw beam diameter before focusing: 22 mm (d_{86})
- Beam-parameter-product: 10 mm x mrad

Protection screen	RP 2000, Lamella laser, No. 4707R0092 Rev. d, aluminium (6060T66) anodised, 1.6/1.5_13.5_100 mm, AlMgSi 0.5 F25 with lamellar seal PP No. 4401R0014 or TPE No. 4101R0037
Specimen size	300 x 300 mm (3 lamellae)
Angle of incidence	90 ° ± 10 ° (lamella centre)
Distance	2F
Laser output	4 kW
Focal spot diameter (computations made with d_{86})	38 mm
Radiation intensity (computations made with d_{86})	$3.57 \cdot 10^6$ W/m^2
Pre-treatment of the test surface (synthetic resin-based spray paint, jet black matt, RAL 9005)	Yes

Protection wall ventilation	Central extraction system in the test cell and a portable weld fume extraction unit at a lateral distance of approx. 30 cm from the specimen (sensing element cross-section 200 x 500 mm)
Indicator	Thermal paper at a distance of 200 mm
Measured values in s (up to penetration of the protection wall)	26, 31, 34, 31, 29, **25**, 28, 31, 28, 28 Lamellar seal PP: 47.4 Lamellar seal TPE: 38.2
Duration of protection (X_{min} · 0.7) according to PL-Info 007	18 s
Duration of protection ((μ − 3σ) · 0.7) according to DIN EN 60825-4	15 s
Remarks	Preliminary power measurement with Primes PocketMonitor: Target 4.1 kW / Actual 3.91 kW Bombardment of the lamella centre results in the more restrictive/most disadvantageous value 

Malte Gomolka
Test engineer